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The K-Theory of Finite Fields, Revisited

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Abstract. A new, short proof is given for the Quillen theorem that calculates the K-theory of finite fields. This proof uses the Gabber rigidity theorem and the homotopy theory of simplicial presheaves.

Key words. Gabber rigidity, simplicial presheaves, finite fields.

1. Introduction

Suppose that p is a prime number, and let $\mathbb{F}_q$ denote the field with $q = p^m$ elements. In general, if L is a field, let $\bar{L}$ denotes its algebraic closure.

The purpose of this note is to present a modern proof of the following well-known result of Quillen [13]:

THEOREM 1. *There are isomorphisms*

$$K_i(\mathbb{F}_q) \cong \begin{cases} \mathbb{Z}/(q^j - 1), & \text{if } i = 2j-1, j > 0, \quad \text{and} \\ 0, & \text{if } i = 2j, j > 0. \end{cases}$$

The basic technical devices that will be used for this proof are the Gabber rigidity theorem [5] and the homotopy theory of simplicial presheaves. The proof also requires the Lang isomorphism.

2. Gabber Rigidity

The form of the Gabber rigidity theorem that will be used throughout this paper is the following:

THEOREM 2 (Gabber). *Suppose that $\mathcal{O}$ is a Henselian local ring with residue field L, and suppose that $1/n \in \mathcal{O}$. Then the residue map $\pi: \mathcal{O} \to L$ induces isomorphisms*

$$\pi_*: K_i(\mathcal{O}, \mathbb{Z}/n) \xrightarrow{\cong} K_i(L, \mathbb{Z}/n)$$

for $i \geqslant 0$.

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